Random Fields on Euclidean Spaces

Adair Neto

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1. Rice's Formula

The Goal

In this first presentation, instead of giving the precise mathematical definitions, I would like to present our goal in an informal way. It won't be rigorous for now, but we will make everything precise later on.

Our main goal is to prove Rice's formula, which is a way of approximating, for large levels $u \in \mathbf{R}$, the **excursion probability**

$$\mathbf{P}(\{\sup_t f(t) \geq u\})$$

Now, let f be a real-valued stochastic process on $[0,T]$ and $u \in \mathbf{R}$. We define the **number of upcrossings** by f of level u in the interval $[0,T]$ by

$$N_u^+(0,T) \triangleq |\{t \in [0,T] : f(t) = u,\ f'(t) > 0\}|$$

The idea here is that the probability that f crosses the level u is close to the probability that there is an upcrossing of u, with some boundary correction term. Since the number of upcrossings of a high level is always small, the probability of an upcrossing is well approximated by the mean number of upcrossings.

How to compute the expected number of upcrossings?

For simplicity, we will assume that $N_u^+(0,T)$ is well-defined and finite. Our first goal is to compute $\mathbf{E}[N_u^+(0,T)]$.

Let δ_x be the Dirac delta function at x, i.e., for any "reasonable" test function g,

$$\int_{\mathbf{R}} \delta_x(y) g(y)\, \mathrm{d}y = g(x)$$

We will also assume that the upcrossing points (i.e., those $t \in [0,T]$ for which $f(t) = u$ and $f'(t) > 0$) are isolated. That is, each one of them can be covered by a small interval I in which there are no other upcrossings and throughout which $f'(t) > 0$.

If we treat δ as a smooth function, by a change of variables we obtain

$$1 = \int_{\mathbf{R}} \delta_x(y)\, \mathrm{d}y = \int_{\mathbf{I}} \delta_x(f(t)) f'(t)\, \mathrm{d}t$$

Concatenating all such intervals I, we get

$$N_u^+(0,T) = \int_0^T \delta_u(f(t)) \mathbf{1}_{[0,\infty]}(f'(t)) f'(t)\, \mathrm{d}t$$

Denoting by p_t the joint density of $(f(t), f'(t))$ and taking expectation,

$$\begin{aligned}\mathbf{E}[N_u^+(0,T)] &= \int_0^T \int_{-\infty}^{\infty} \int_0^{\infty} y\delta_u(x) p_t(x,y)\, \mathrm{d}y\, \mathrm{d}x\, \mathrm{d}t \\ &= \int_0^T \int_0^{\infty} y p_t(u,y)\, \mathrm{d}y\, \mathrm{d}t\end{aligned}$$

This is the basic form of **Rice's formula**. However, in practice it could be difficult to compute this integral.

Let us assume that f is a standard Gaussian. In this case, $f(t)$ and $f'(t)$ are independent for each t (notice that $\mathbf{E}[f(t)f_j(t)]$ equals the partial derivative in the t_j-direction of the covariance, which is constant).

Denoting the variance of $f'(t)$ by λ_t, we have

$$\begin{aligned}\mathbf{E}[N_u^+(0,T)] &= \frac{e^{-u^2/2}}{2\pi} \int_0^T \int_0^{\infty} y \frac{e^{-\frac{y^2}{2\lambda_t}}}{\sqrt{\lambda_t}}\, \mathrm{d}y\, \mathrm{d}t \\ &= \frac{e^{-u^2/2}}{2\pi} \int_0^T \lambda_t^{1/2}\, \mathrm{d}t\end{aligned}$$

Now, if we assume further that f is stationary, then λ_t equals something called the *second spectral moment* λ_2. In this case, we obtain the **classic Rice's formula**

$$\mathbf{E}[N_u^+(0,T)] = \frac{\lambda_2^{1/2} T}{2\pi} e^{-u^2/2}$$

How is this related to excursion sets?

The interesting thing here is that Rice's formula is related to **excursion sets** $A_u \triangleq \{t \in T : f(t) \geq u\}$ and their Euler characteristic $\chi(A_u)$. We will talk a bit about the Euler characteristic later on, but for a "nice" set $A \subseteq \mathbf{R}$, $\chi(A)$ is just the number of connected components of A. Then it is immediate that

$$\chi(A_u) = \mathbf{1}_{\{f_0 \geq u\}} + N_u^+$$

which implies

$$\begin{aligned}\mathbf{E}[\chi(A_u)] &= \mathbf{P}[f_0 \geq u] + \mathbf{E}[N_u^+] \\ &= \Psi(u) + \frac{T\lambda_2^{1/2}}{2\pi} e^{-u^2/2}\end{aligned}$$

where $\Psi \triangleq 1 - \Phi$ and Φ is the CDF of the standard normal distribution.

The idea here is that increasing λ_2 increases the "high-frequency" components of f, i.e., the sample paths become rougher.

Our goal for the next sessions is to extend the last equation from processes on $\mathbf{R}$ to random fields on $\mathbf{R}^d$ in the stationary case. For the nonstationary case, it is necessary to work with manifolds.

2. What is a random field?

Our goal is to study the geometry of the excursion sets of smooth, $\mathbf{R}^d$-valued random fields f over piecewise smooth manifolds M.

Excursion sets are subsets of M given by

$$A_D = A_D(f, M) \triangleq \{t \in M : f(t) \in D\}, \quad D \subseteq \mathbf{R}^d$$

or

$$A_u = A_u(f, T) \triangleq \{t \in T : f(t) \geq u\}$$



Fig. 11.1.1. Realizations of Gaussian sample paths with the spectrum (11.1.7) and second spectral moments $\lambda_2 = 200$ (left) and $\lambda_2 = 1{,}000$ (right).

Figure 1: Sample Paths

Definitions

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a complete (i.e., every subset of every null set is measurable) probability space, and T a topological space. If you don't know what a topological space is, just think of $T \subseteq \mathbf{R}^N$.

Recall that a random process is a family $\{X_t : t \in T\}$ of random variables indexed by "time", either discrete ($T = \mathbf{N}_0$) or continuous ($T = [0, \infty)$). A random field is a a generalisation of this notion, changing from time to something more general, e.g., the space $\mathbf{R}^d$ or any topological space T.

More precisely, a **random field** is measurable mapping $f : \Omega \longrightarrow (\mathbf{R}^T)^d$.

If $d = 1$, it is a real-valued random field, and if $d > 1$, then it is a vector-valued random field.

If $T \subseteq \mathbf{R}^N$, we call f an (N, d) random field. If $d = 1$, we call it an N-dimensional random field.

A real-valued **Gaussian field** is a random field for which all finite-dimensional distributions of $(f_{t_1}, \ldots, f_{t_n})$ are multivariate Gaussian for each $1 \leq n < \infty$ and $(t_1, \ldots, t_n) \in T^n$.

We also define:

- **Mean:** $m(t) \triangleq \mathbf{E}[f(t)]$;
- **Covariance:** $C(s,t) \triangleq \mathbf{E}[(f_s - m_s)(f_t - m_t)]$;
- A (pseudo)metric on T by
$$d(s,t) \triangleq \sqrt{\mathbf{E}[(f(s) - f(t))^2]}.$$

Random fields have many applications in image analysis, spatial statistics, rainfall data, brain mapping etc. These won't be discussed here.

Brownian Sheet

The **Brownian sheet** $W = \{W(t) : t \in \mathbf{R}_+^N\}$ is a centred (N, d) Gaussian field whose covariance function is

$$\mathbf{E}[W_i(s) W_j(t)] = \delta_{ij} \prod_{k=1}^{N} (s_k \wedge t_k)$$

Notice that for $N = 1$, W is a Brownian motion in $\mathbf{R}^d$.

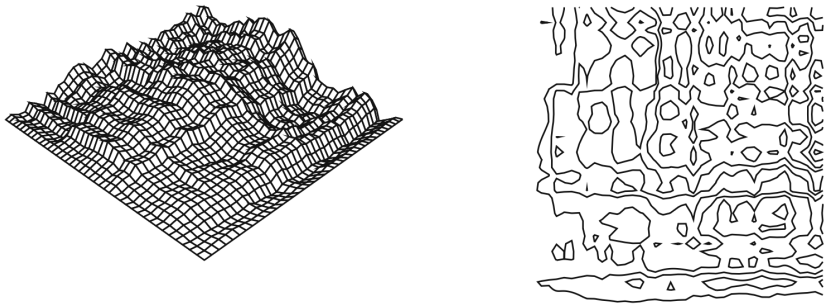


Fig. 1.4.1. A simulated Brownian sheet on $[0, 1]^2$, along with its contour lines at the zero level.

Figure 2: Brownian Sheet

Stationary Fields

A random field is said to be **left-stationary** if its finite-dimensional distributions are invariant under left-multiplication, i.e., for any $n \geq 1$, $t_1, \ldots, t_n \in T$ and $g \in G$ (a group acting on T), we have

$$(f(t_1), \ldots, f(t_n)) \stackrel{d}{=} (f(gt_1), \ldots, f(gt_n))$$

Notice that if f is stationary and centred (i.e., $\mathbf{E}[f_t] = 0$), we may write $C(s,t) = C(t-s)$. In fact,

$$C(s,t) = \mathbf{E}[f(s)f(t)] = \mathbf{E}[f(0)f(s^{-1}t)] = C(t-s)$$

Suppose that f is a complex-valued field, stationary, centred, and $T = \mathbf{R}^N$.

Since f is stationary, we are allowed to write

$$C(s,t) = \mathbf{E}[f_s \overline{f_t}] = C(t-s)$$

Spectral Distribution Theorem

We will use the following result, which is called the **Spectral Distribution Theorem** or **Bochner's Theorem** (see Doob's Stochastic Processes, Theorem 3.2).

A continuous function $C : \mathbf{R}^N \longrightarrow \mathbf{C}$ is nonnegative definite (i.e., a covariance function) if, and only if, there exists a finite measure ν on the Borel σ-algebra $\mathcal{B}(\mathbf{R}^N)$ such that

$$C(t) = \int_{\mathbf{R}^N} e^{i\langle t,\lambda\rangle}\, \nu(\mathrm{d}\lambda), \quad \forall\, t \in \mathbf{R}^N$$

We write $\sigma^2 \triangleq C(0) = \nu(\mathbf{R}^N)$.

The measure ν above is called the **spectral measure** for the covariance function C, and the function $F : \mathbf{R}^N \longrightarrow [0, \sigma^2]$ given by

$$F(\lambda) = \nu\left(\prod_{i=1}^{N} (-\infty, \lambda_i]\right), \quad \lambda = (\lambda_1, \ldots, \lambda_N) \in \mathbf{R}^N$$

is called the **spectral distribution function**.

When F is absolutely continuous (i.e., F has a derivative a.e., the derivative is integrable, and $\int_a^x F'(t)\ \mathrm{d}t = F(x) - F(a)$), the corresponding density is called the **spectral density**.

Given the spectral representation above, we define the **spectral moments**

$$\lambda_{i_1,\ldots,i_N} \triangleq \int_{\mathbf{R}^N} \lambda_1^{i_1} \cdots \lambda_N^{i_N}\ \nu(\mathrm{d}\lambda)$$

for all $(i_1, \ldots, i_N)$ with $i_j \geq 0$.

Isotropic Fields

An important class of random fields is when every direction "looks the same". This gives rise to the following definition.

An **isotropic field** on $\mathbf{R}^N$ is a stationary random field with the property that the covariance function depends only on the Euclidean length of the vector t, i.e,

$$C(t) = C(|t|)$$

In an isotropic field, the spectral measure is spherically symmetric. In one line, this follows from the spectral distribution theorem and the fact that rotations are orthogonal transformations. If you want to be more detailed,

- Let $\theta : \mathbf{R}^N \longrightarrow \mathbf{R}^N$ be a rotation.
- Since θ is a rotation, $|\theta(t)| = |t|$.
- By the definition of isotropy, $C(t) = C(\theta(t))$.
- Applying the spectral distribution theorem,

$$\begin{aligned}\int_{\mathbf{R}^N} e^{i\langle t,\lambda\rangle}\ \nu(\mathrm{d}\lambda) &= \int_{\mathbf{R}^N} e^{i\langle \theta(t),\lambda\rangle}\ \nu(\mathrm{d}\lambda) \\ &= \int_{\mathbf{R}^N} e^{i\langle t,\theta^{-1}(\lambda)\rangle}\ \nu(\mathrm{d}\lambda) \\ &= \int_{\mathbf{R}^N} e^{i\langle t,\lambda\rangle}\ \nu_\theta(\mathrm{d}\lambda)\end{aligned}$$

  where ν_θ is the push-forward $\nu_\theta(A) = \nu(\theta^{-1}A)$.
- Since this holds for all t, $\nu \equiv \nu_\theta$.

This fact has two important consequences:

1. An isotropic field cannot have all the probability of its spectral measure concentrated in one small region away from the origin.
2. It simplifies the structure of the spectral moments. It is possible to prove that (see Adler and Taylor's Section 5.5)
$$\mathbf{E}[f_i(t)f_j(t)] = -\mathbf{E}[f(t)f_{ij}(t)] = \delta_{ij}\lambda_2$$
where λ_2 is the **second spectral moment**
$$\lambda_2 \triangleq \int_{\mathbf{R}^N} \lambda_i^2\ \nu(\mathrm{d}\lambda)$$
which is independent of the value of i, since the field is isotropic and ν is rotation-invariant.

3. Reproducing Kernel Hilbert Spaces

The main goal of this part is to show that every centred Gaussian process with a continuous covariance function has an expansion of the form

$$f(t) = \sum_{n=1}^{\infty} \xi_n \varphi_n(t)$$

where ξ_n are i.i.d. standard normal variables, and φ_n are functions of T determined by the covariance function C.

This convergence is, in general, in $L^2(\mathbf{P})$ for each $t \in T$. If f is a.s. continuous, then the convergence is uniform over T almost surely.

A striking consequence of this expansion is that, if the covariance is continuous, then pointwise and global a.s. continuity are equivalent (for Gaussian processes), i.e.,

$$\mathbf{P}\left(\lim_{s\to t} f_s = f_t, \forall\, t \in T\right) = 1 \iff \mathbf{P}\left(\lim_{s\to t} f_s = f_t\right) = 1, \forall\, t \in T$$

The key to this construction is something called "Reproducing Kernel Hilbert Spaces", which is widely used in Machine Learning. So, let us recall the definition of a Hilbert space.

A **Hilbert Space** is a vector space equipped with an inner product which is complete (i.e., all Cauchy sequences converge to an element in the space). The most important example is $L^2(\mu)$, which is the space of square-integrable functions with inner-product given by

$$\langle f, g\rangle \triangleq \int fg \, \mathrm{d}\mu$$

Reproducing Kernel Property

Let T be a compact space, f a centred Gaussian with covariance function C, and

$$S \triangleq \left\{u : T \longrightarrow \mathbf{R} : u(\cdot) = \sum_{i=1}^{n} a_i C(s_i, \cdot),\ a_i \in \mathbf{R}, s_i \in T, n \geq 1\right\}$$

We define an inner product on S by setting

$$\langle u, v\rangle_S = \left\langle \sum_{i=1}^{n} a_i C(s_i, \cdot), \sum_{j=1}^{m} b_j C(t_j, \cdot)\right\rangle_S = \sum_{i=1}^{n}\sum_{j=1}^{m} a_i b_j C(s_i, t_j)$$

Since C is non-negative definite, we have $\langle u, u\rangle_S \geq 0$ for all $u \in S$.

And we have the **reproducing Kernel property**,

$$\langle u, C(t,\cdot)\rangle_S = \left\langle \sum_{i=1}^{n} a_i C(s_i,\cdot), C(t,\cdot)\right\rangle_S = \sum_{i=1}^{n} a_i C(s_i, t) = u(t)$$

Let us assume that C is positive definite. Then $\langle u, u\rangle = 0$ if, and only if, $u \equiv 0$. In this case, $\langle\cdot,\cdot\rangle_S$ induces a norm $\|\cdot\|_S$.

If $\{u_n\}$ is a sequence in S, then by the reproducing Kernel property and Cauchy-Schwarz inequality,

$$\begin{aligned} |u_n(t) - u_m(t)| &= |\langle u_n - u_m, C(t,\cdot)\rangle_S| \\ &\leq \|u_n - u_m\|_S \|C(t,\cdot)\|_S \\ &\leq \|u_n - u_m\|_S C(t,t) \end{aligned}$$

That is, if $\{u_n\}$ is Cauchy in $\|\cdot\|_S$, then it is pointwise Cauchy.

The closure of S under $\|\cdot\|_S$, denoted by $H(C)$ is called the **Reproducing Kernel Hilbert Space** (RKHS).

Notice that $H(C)$ is separable, since T is compact (every compact space is separable) and C is continuous (continuous image of separable is separable). Wherefore, every $u \in H(C)$ satisfies the reproducing Kernel property.

Trivial Example

Consider - $T = \{1, \ldots, n\}$; - f a centred Gaussian with covariance matrix given by $C = (c_{ij})$, $c_{ij} = \mathbf{E}[f_i f_j]$; - $C^{-1} = (c^{ij})$ the inverse of C, which exists by positive definiteness.

Then the reproducing kernel Hilbert space of f consists of all n-dimensional vectors with inner product

$$\langle u, v\rangle_S = \sum_{i=1}^{n}\sum_{j=1}^{n} u_i c^{ij} v_j$$

In fact, denoting by k the kth row of C,

$$\langle u, C_k\rangle_S = \sum_{i=1}^{n}\sum_{j=1}^{n} u_i c^{ij} c_{kj} = \sum_{i=1}^{n} u_i \delta_{i,k} = u_k$$

Example: Cameron-Martin space

Let $f = W$ be the standard Brownian motion on the unit interval $T = [0, 1]$. Recall that the covariance is given by $C(s, t) = s \wedge t$.

Take two elements in S

$$u(t) = \sum_{i=1}^{n} a_i C(s_i, t), \quad v(t) = \sum_{j=1}^{n} b_j C(t_j, t)$$

with inner product

$$\langle u, v\rangle_S = \sum_{i=1}^{n}\sum_{j=1}^{n} a_i b_j \min\{s_i, t_j\}$$

Since the derivative of $C(s, t)$ with respect to t is $\mathbf{1}_{[0,s]}(t)$, the derivative of u is

$$u'(t) = \sum_{i=1}^{n} a_i \mathbf{1}_{[0,s_i]}(t)$$

Hence,

$$\begin{aligned}\langle u, v\rangle_S &= \sum_{i=1}^{n}\sum_{j=1}^{n} a_i b_j \int_0^1 \mathbf{1}_{[0,s_i]}(t)\mathbf{1}_{[0,t_j]}(t)\,\mathrm{d}t \\ &= \int_0^1 \left(\sum_{i=1}^{n} a_i \mathbf{1}_{[0,s_i]}(t) \sum_{j=1}^{n} b_j \mathbf{1}_{[0,t_j]}(t)\right)\mathrm{d}t \\ &= \int_0^1 u'(t)v'(t)\,\mathrm{d}t\end{aligned}$$

This motivates us to define

$$H \triangleq \left\{u : u(t) = \int_0^t u'(s)\,\mathrm{d}s,\ \int_0^t (u'(s))^2\,\mathrm{d}s < \infty\right\}$$

and equip this space with the inner product

$$\langle u, v\rangle_H \triangleq \int_0^1 u'(s)v'(s)\,\mathrm{d}s$$

Notice that $C(s,\cdot) \in H$ for $t \in [0,1]$, and

$$\langle u, C(t,\cdot)\rangle_H = \int_0^1 u'(s)\mathbf{1}_{[0,t]}(s)\,\mathrm{d}s = u(t)$$

This proves that our candidate H is indeed our RKHS. It is known as a **Cameron-Martin space**.

4. Orthogonal Expansions

Countable orthonormal basis for the Hilbert space

We start by showing a (canonical) unitary transformation (that is, an isometric isomorphism) Θ between $H(C)$ and

$$\mathcal{H} \triangleq \overline{\mathrm{span}\{f_t : t \in T\}}$$

with the closure in $L^2(\mathbf{P})$ and the (covariance) inner product inherited as a subspace of $L^2(\mathbf{P})$. Indeed, we define the linear mapping $\Theta : S \longrightarrow \mathcal{H}$ by

$$\Theta(u) = \Theta\left(\sum_{i=1}^n a_i C(t_i,\cdot)\right) \triangleq \sum_{i=1}^n a_i f(t_i)$$

Notice that $\Theta(u)$ is Gaussian for each $u \in S$, Θ preserves the norm:

$$\left\|\Theta\left(\sum_{i=1}^n a_i C(t_i,\cdot)\right)\right\|_H^2 = \sum_{i,j} a_i C(t_i,t_j)a_j = \left\|\sum_{i=1}^n a_i f(t_i)\right\|_2^2,$$

and it is possible to extend Θ to $H(C)$ with range equal to $\mathcal{H}$. In fact, given $u \in H(C)$, choose a sequence $(u_n) \subseteq S$ such that $u_n \to u$ in $H(C)$. Since Θ is an isometry,

$$\|\Theta(u_n) - \Theta(u_m)\|_{L^2(\mathbf{P})} = \|u_n - u_m\|_{H(C)}.$$

Hence, $(\Theta(u_n))$ is a Cauchy sequence in $L^2(\mathbf{P})$, and therefore converges to some limit (by the completeness of the space), which we call

$$\tilde{\Theta}(u) \triangleq \lim_{n\to\infty} \Theta(u_n).$$

Notice that this is well-defined (i.e., does not depend on the choice of the approximating sequence). If $(v_n) \subseteq S$ is another sequence with $v_n \to u$ in $H(C)$, then (and we drop the tilde for simplicity)

$$\|\Theta(u_n) - \Theta(v_n)\|_2 = \|u_n - v_n\|_{H(C)} \to 0.$$

Since we have an isometric isomorphism and $H(C)$ is separable, it follows that $\mathcal{H}$ is also separable.

Now, given an orthonormal basis $\{\varphi_n\}_{n\geq 1}$ for $H(C)$, setting $\xi_n \triangleq \Theta(\varphi_n)$ gives an orthonormal basis $\{\xi_n\}$ for $\mathcal{H}$. Furthermore, the ξ_n are standard normal variables, and

$$f_t = \sum_{n=1}^\infty \xi_n \langle f_t, \xi_n\rangle_{\mathcal{H}} = \sum_{n=1}^\infty \xi_n \mathbf{E}[f_t \xi_n]$$

with convergence in $L^2(\mathbf{P})$.

Using the fact that Θ is an isometry and the reproducing kernel property, we know that

$$\mathbf{E}[f_t \xi_n] = \langle \Theta(C(t,\cdot)), \Theta(\varphi_n) \rangle_{\mathcal{H}} = \langle C(t,\cdot), \varphi_n \rangle_S = \varphi_n(t)$$

Combining the last two equations, we have proven the following.

[!theorem] The random field f has the representation

$$f_t = \sum_{n=1}^{\infty} \xi_n \varphi_n(t)$$

with convergence in $L^2(\mathbf{P})$.

A stronger version of this result

Our goal now is to prove:

[!theorem] If f is a.s. continuous, then the series above converges uniformly on T a.s.

To prove this result, we need two lemmas. The first one is a generalisation of Lévy's theorem (see Section 18.1 in Williams' Probability with Martingales).

> [!theorem] Itô-Nisio Theorem Let $(Z_n)_{n\geq 0}$ be a sequence of symmetric independent random variables taking values in a separable, real Banach space B and let $X_n \triangleq \sum_{i=1}^{n} Z_i$. Then X_n converges a.s. if, and only if, there exists a B-valued random variable X such that $\langle X_n, x^* \rangle \longrightarrow \langle X, x^* \rangle$ in probability for every $x^* \in B^*$.

For a proof, see Theorem 2.4 in Ledoux and Talagrand's Probability in Banach Spaces.

> [!lemma] Let $\{\varphi_n\}_{n\geq 1}$ be an orthonormal basis for $H(C)$ and assume that the covariance function is continuous. Then each φ_n is continuous and
>
> $$\sum_{n=1}^{\infty} \varphi_n^2(t)$$
>
> converges uniformly in $t \in T$ to $C(t,t)$.

Notice that, using the reproducing kernel property and Cauchy-Schwarz,

$$\begin{aligned}
|\varphi_n(s) - \varphi_n(t)| &= |\langle C(s,\cdot), \varphi_n(\cdot) \rangle_S - \langle C(t,\cdot), \varphi_n(\cdot) \rangle_S| \\
&= |\langle [C(s,\cdot) - C(t,\cdot)], \varphi_n(\cdot) \rangle_S| \\
&\leq \|\varphi_n\|_S \|C(s,\cdot) - C(t,\cdot)\|_S \\
&= \|C(s,\cdot) - C(t,\cdot)\|_S \\
&= C(s,s) - 2C(s,t) + C(t,t) \\
&= d^2(s,t)
\end{aligned}$$

This proves the continuity claim. To prove the uniform convergence, we use the orthonormal expansion and the reproducing kernel property to obtain

$$C(t,\cdot) = \sum_{n=1}^{\infty} \varphi_n(\cdot) \langle C(t,\cdot), \varphi_n \rangle_S = \sum_{n=1}^{\infty} \varphi_n(\cdot) \varphi_n(t)$$

where the convergence is in the $\|\cdot\|_S$ norm. Thus, $\sum_{n=1}^{\infty} \varphi_n^2(t)$ converges to $C(t,t)$ for every $t \in T$.

Since the convergence is monotone, by Dini's theorem (compact plus monotone convergence implies uniform convergence), it is also uniform. This finishes the proof of the lemma.

Using these results, let us sketch how to conclude the proof of our goal.

1. Apply Itô-Nisio Theorem to conclude that $\sum_{n=1}^{\infty} \xi_n \varphi_n(t)$ converges a.s. to a limit f_t, which is almost surely continuous by assumption.
2. Consider f and $\xi_n \varphi_n$ as random variables in the Banach space $C^0(T)$ with sup-norm topology. Then elements of the dual $C^*(T)$ are finite, signed Borel measures μ on T, and $\langle f, \mu \rangle = \int f \, \mathrm{d}\mu$.
3. Define
$$f_n(\cdot) = \sum_{i=1}^{n} \xi_i \varphi_i(\cdot) = \sum_{i=1}^{n} \Theta(\varphi_i) \varphi_i(\cdot)$$
4. Using Itô-Nisio again, one can check using the lemma above that, for every $\mu \in C^*(T)$, the random variables $\langle f_n, \mu \rangle$ converge in probability to $\langle f, \mu \rangle$.

Karhunen-Loève Expansion

How to find the orthonormal basis $\{\varphi_n\}$? This is doable when $T \subset \mathbf{R}^N$.

Suppose that $T = [0,1]^N$ and consider the following operator

$$\begin{aligned} C &: L^2(T) \longrightarrow L^2(T), \\ (C\psi)(t) &= \int_T C(s,t)\psi(s) \, \mathrm{d}s. \end{aligned}$$

Since C is continuous, symmetric, and positive definite on the compact set T, the map C with kernel C is Hilbert-Schmidt, thus compact. Symmetry of the covariance function implies that C is self-adjoint, and positivity of C implies that C is positive. By the [[Spectral theorem for compact self-adjoint operators]], there exists eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots$ (with $\lambda_n \downarrow 0$ and accumulating only at zero) and a normalised family of eigenfunctions $\{\psi_n\}$.

Then Mercer's Theorem (see Theorem 11.6.7 in Dieudonné's Foundation of Modern Analysis) states that (for a positive compact operator defined by an Hermitian kernel)

$$C(s,t) = \sum_{n=1}^{\infty} \lambda_n \psi_n(s) \psi_n(t)$$

where the series converges absolutely and uniformly on $[0,1]^N \times [0,1]^N$.

[!Lemma] Let f be as above. Then $\{\sqrt{\lambda_n}\psi_n\}$ is a complete orthonormal system in $H(C)$.

The proof is as follows.

1. Set $\varphi_n \triangleq \sqrt{\lambda_n}\psi_n$.
2. Define
$$H \triangleq \left\{ h : h(t) = \sum_{n=1}^{\infty} a_n \varphi_n(t), \ t \in [0,1]^N, \ \sum_{n=1}^{\infty} a_n^2 < \infty \right\}$$
3. Equip H with the inner product
$$\langle h, g \rangle_H = \left\langle \sum_{n=1}^{\infty} a_n \varphi_n, \sum_{n=1}^{\infty} b_n \varphi_n \right\rangle_H \triangleq \sum_{n=1}^{\infty} a_n b_n$$
4. Check that H has the reproducing kernel property:
$$\begin{aligned} \langle h(\cdot), C(t,\cdot) \rangle_H &= \left\langle \sum_{n=1}^{\infty} a_n \varphi_n, \sum_{n=1}^{\infty} \sqrt{\lambda_n} \psi_n(t) \varphi_n(\cdot) \right\rangle_H \\ &= \sum_{n=1}^{\infty} \sqrt{\lambda_n} a_n \psi_n(t) = h(t) \end{aligned}$$

5. Hence, H is a RKHS with reproducing kernel C and $\{\sqrt{\lambda_n}\psi_n\}$ is an orthonormal system for $H(C)$. In fact, it is a basis by construction of H and, by the definition of the inner product,

$$\langle \varphi_n, \varphi_m \rangle_H = \left\langle \sum_k \delta_{kn}\varphi_k, \sum_k \delta_{km}\varphi_k \right\rangle_H = \sum_k \delta_{kn}\delta_{km} = \delta_{nm}.$$

Thus $\{\varphi_n\}$ is an orthonormal system in H.

To obtain the Karhunen-Loève Expansion, we just need to set $\varphi_n = \sqrt{\lambda_n}\psi_n$ in our previous orthogonal expansion and then

$$f(t) = \sum_{n=1}^{\infty} \sqrt{\lambda_n}\xi_n\psi_n(t).$$

Example: The Karhunen-Loève Expansion of the Brownian Sheet

Let us consider again the standard Brownian motion on $[0, 1]$. In this case, the eigenvalue equation for our operator C becomes

$$\begin{aligned}\lambda\psi(t) &= \int_0^1 \min(s, t)\psi(s)\, \mathrm{d}s \\ &= \int_0^t s\psi(s)\, \mathrm{d}s + t\int_t^1 \psi(s)\, \mathrm{d}s\end{aligned}$$

Differentiating with respect to t yields

$$\lambda\psi'(t) = \int_t^1 \psi(s)\, \mathrm{d}s$$

and

$$\lambda\psi''(t) = -\psi(t)$$

Remark that $\psi(0) = 0$ and $\psi'(1) = 0$.

Solving these differential equations, we obtain

$$\psi_n(t) = \sqrt{2}\sin\left(\frac{1}{2}(2n+1)\pi t\right), \quad \lambda_n = \left(\frac{2}{(2n+1)\pi}\right)^2$$

Thus, the Karhunen-Loève expansion is given by

$$W_t = \frac{\sqrt{2}}{\pi}\sum_{n=0}^{\infty} \xi_n \left(\frac{2}{2n+1}\right)\sin\left(\frac{1}{2}(2n+1)\pi t\right)$$

Recall that the Brownian sheet has the covariance function $\mathbf{E}[W_s W_t] = (s_1 \wedge t_1) \times \cdots \times (s_N \wedge t_N)$. In this case, using a multi-index $n = (n_1, \ldots, n_N)$, it follows that the eigenpair is given by

$$\psi_n(t) = 2^{N/2}\prod_{i=1}^{N} \sin\left(\frac{1}{2}(2n_i+1)\pi t_i\right), \quad \lambda_n = \prod_{i=1}^{N}\left(\frac{2}{(2n_i+1)\pi}\right)^2$$

Appendix

The Euler Characteristic

Heuristically, the **Euler Characteristic** is the unique integer-valued map χ defined on "nice sets" A such that

$$\chi(A) = \begin{cases} 0, & \text{if } A = \emptyset \\ 1, & \text{if } A \text{ is 'ball-like'} \end{cases}$$

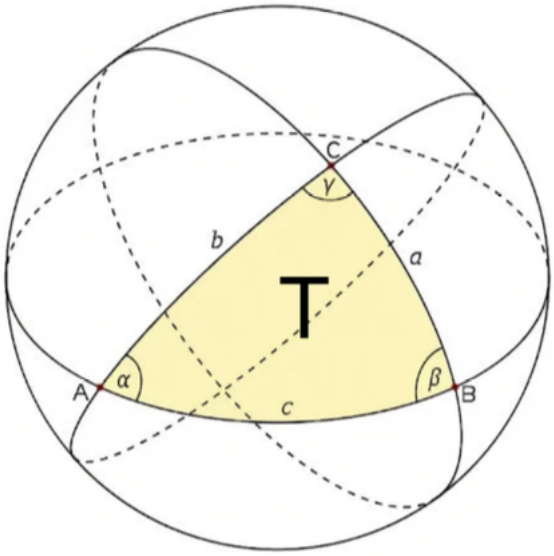
where 'ball-like' means homotopically equivalent to the N-ball, and with the additivity property

$$\chi(A \cup B) = \chi(A) + \chi(B) - \chi(A \cap B)$$

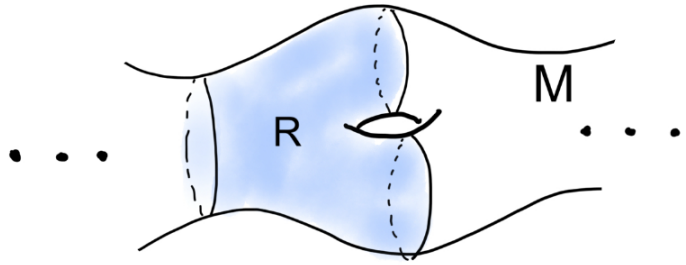
In $\mathbf{R}^2$, $\chi(A)$ is the number of connected components of A minus the number of 'holes'.

In $\mathbf{R}^3$, it is the number of connected components, minus the number of 'handles', plus the number of holes.

If you think of a surface in $\mathbf{R}^3$, the idea is to triangulate a 'regular' region and then define $\chi(R) = V - E + F$.



As an example, one can use the Gauss-Bonnet theorem to show that the pair of pants R has Euler characteristic $\chi(R) = -1$.



More generally, it is an alternating sum of the dimensions of something called the 'cohomology space'. But we won't enter into that here.

Mean Euler Characteristic

Setting

Assume that

- f is a centred, stationary Gaussian field on a rectangle $T \subseteq \mathbf{R}^N$;

- For each $t \in T$, the joint distribution of $(\partial_i f(t), \partial^2_{ij} f(t))_{i,j=1,\dots,N}$ is nondegenerate (i.e., the Hessian has nonzero determinant);
- And there exists a constant K such that, for all $s, t \in T$,

$$\max_{i,j} |C_{\partial^2_{ij} f}(t,t) + C_{\partial^2_{ij} f}(s,s) - 2C_{\partial^2_{ij} f}(s,t)| \le K|\log|t-s||^{-(1+\alpha)}$$

  for some $\alpha > 0$.

We write the second-order spectral moments

$$\lambda_{ij} = \int_{\mathbf{R}^N} \lambda_i \lambda_j \ \nu(\mathrm{d}\lambda)$$

and $\Lambda \triangleq (\lambda_{ij})$ for the $N \times N$ matrix of these moments.

In this case, it is possible to prove (see Section 5.5 of Adler and Taylor's Random Fields and Geometry) that

$$\mathbf{E}[f_i(t) f_j(t)] = \lambda_{ij} = -\partial^2_{ij} C(0)$$

i.e., Λ is the covariance matrix of ∇f.

Let us fix the following:

- $T = \prod_{i=1}^N [0, T_i]$ (a rectangle in $\mathbf{R}^N$);
- $\mathcal{J}_k = \mathcal{J}_k(T)$ be the collection of $2^{N-k}\binom{N}{k}$ faces of dimension k in T, which we take to be closed;
- O_k denotes the $\binom{N}{k}$ elements of $\mathcal{J}_k$ which include the origin;
- Fix $J \in \mathcal{J}_k$. We write Λ_J for the $k \times k$ matrix with elements λ_{ij}, $i, j \in \sigma(J)$, where $\sigma(J)$ is a subset of $\{1, \dots, N\}$ of size k;
- Let $\Psi \triangleq 1 - \Phi$ denote the tail probability function of a standard normal variable;
- Denote by H_n the n-th Hermite polynomial

$$H_n(x) \triangleq (-1)^n e^{\frac{x^2}{2}} \frac{d^n}{dx^n} e^{-\frac{x^2}{2}} = n! \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(-1)^j x^{n-2j}}{j!(n-2j)!2^j}, \quad x \in \mathbf{R}.$$

Main Theorem

In this context, we have that

$$\mathbf{E}[\chi(A_u)] = e^{-\frac{u^2}{2\sigma^2}} \sum_{k=1}^N \sum_{J \in O_k} \frac{|J||\Lambda_J|^{1/2}}{(2\pi)^{(k+1)/2}\sigma^k} H_{k-1}\left(\frac{u}{\sigma}\right) + \Psi\left(\frac{u}{\sigma}\right)$$

Notice that the k-dimensional volume $|J|$ is given by

$$|J| = \prod_{i \in \sigma(J)} T_i$$

Corollary: the isotropic case

An important corollary is the isotropic case. If f is isotropic and T is the cube $[0, T]^N$, and denoting by λ_2 the variance of $\partial_i f$ (which is invariant of i by isotropy), then

$$\mathbf{E}[\chi(A_u)] = e^{-\frac{u^2}{2\sigma^2}} \sum_{k=1}^N \frac{\binom{N}{k} T^k \lambda_2^{k/2}}{(2\pi)^{(k+1)/2}\sigma^k} H_{k-1}\left(\frac{u}{\sigma}\right) + \Psi\left(\frac{u}{\sigma}\right)$$

This result follows from the spherical symmetry of the spectral measure (recall that $\mathbf{E}[f_i(t) f_j(t)] = -\mathbf{E}[f(t) f_{ij}(t)] = \delta_{ij}\lambda_2$), which implies that each matrix Λ_J is equal to $\lambda_2 I$.

Example: Rice's formula

If $N = 1$, then T is simply the interval $[0, T]$.

By definition of Hermite polynomials, $H_0 = 1$ and we obtain

$$\mathbf{E}[\chi(A_u)] = \Psi\left(\frac{u}{\sigma}\right) + \frac{T\lambda_2^{1/2}}{2\pi\sigma} e^{-\frac{u^2}{2\sigma^2}}$$

thus recovering Rice's formula.

Notice that, as $u \to -\infty$, $\mathbf{E}[\chi(A_u)] \to 1$. The geometric intuition behind this is that once $u < \inf_T f(t)$, we have $A_u \equiv T$ and $\chi(A_u) = \chi(T)$. This is a general phenomenon, independent of dimension of the topology of T.

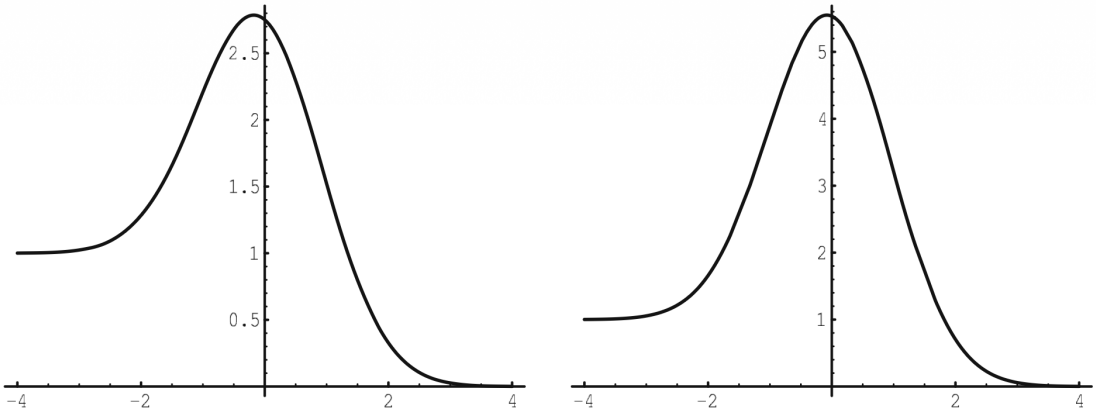


Fig. 11.7.1. $\mathbb{E}\{\varphi(A_u)\} : N = 1.$

Example: dimension two

If $N = 2$ and $\sigma = 1$, since $H_1(x) = x$,

$$\mathbf{E}[\chi(A_u)] = \Psi(u) + \left[\frac{T^2\lambda_2}{(2\pi)^{3/2}} u + \frac{2T\lambda_2^{1/2}}{2\pi}\right] e^{-\frac{u^2}{2}}$$

The negative values are due to the excursion sets having, in the mean, more holes than connected compon-

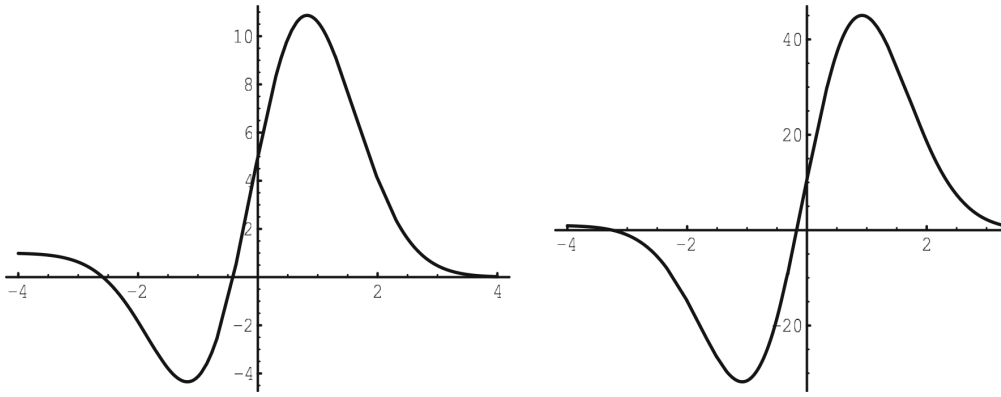


Fig. 11.7.2. $\mathbb{E}\{\varphi(A_u)\} : N = 2.$

ents for most negative values of u.